

Custom Bubble Cursor Pointing Interaction Study

Introduction & Motivation:

Although the traditional mouse pointer remains the standard for graphical user interface interaction, alternative pointing techniques such as the Bubble Cursor offer a unique way to select that may improve speed and accuracy. The Bubble Cursor dynamically adjusts its size to snap to the nearest selectable target, reducing the required precision and effort for target acquisition. For my final project, I designed a custom bubble cursor that builds on the existing implementation to study and compare user performance when used against traditional mouse pointer. Using Fitts's Law metrics, I evaluated selection time, accuracy, and efficiency across both techniques.

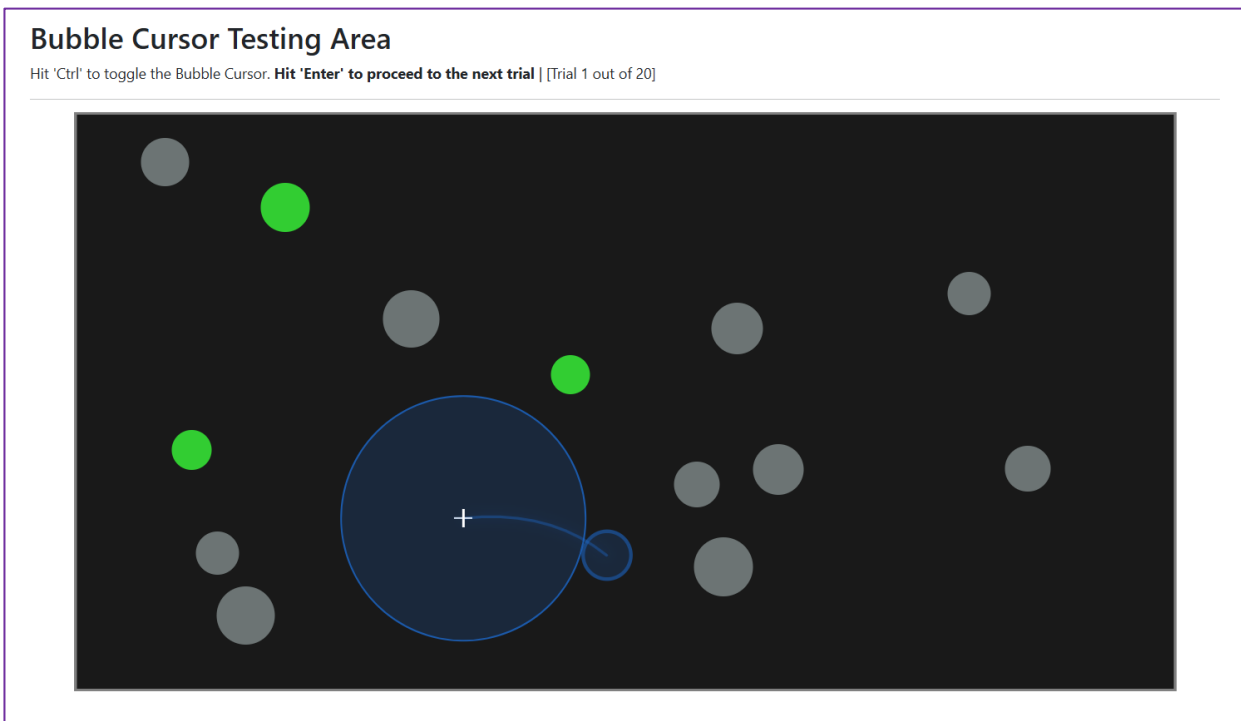
Design & Implementation:

The system was built using a lightweight web stack consisting of HTML, CSS, and JavaScript, using Node.js and Express to serve the **training** and **test trial pages**. My goal was to create an intuitive, visually supportive pointing interaction platform that enables the seamless use of my custom Bubble Cursor's against a traditional mouse. The Bubble Cursor itself was represented by a **blue ring** with a solid border, optimized for visibility against a **dark background**. The ring dynamically grows or shrinks based on proximity to nearby selectable items, making it easier for users to acquire targets without the need for pixel-precise aiming.

At the core of my cursor's functionality is the *getDistanceToNearestTarget()* function, which calculates the distance from the cursor to the center of all on-screen targets and dynamically resizes the bubble so that it touches the edge of the nearest item/target, allowing the user to select it without hovering over the item/target. This behavior effectively reduces the index of difficulty, ideally when dealing smaller or closely packed targets.

A key implementation feature that separates this bubble cursor from the original implementation was the combined use of the dynamic attractor curve and automatic target highlighting. As the user moves the bubble cursor, a **smooth blue curve** is drawn from the center of the bubble to the center of the nearest target, gently guiding the user's attention without interfering with cursor movement. At the same time, the closest target was **automatically shaded** light blue to visually reinforce which item was currently selectable. Together, these elements provided continuous but subtle feedback to the user. Additionally, the Bubble Cursor could be toggled on or off at any time using the **CTRL** key—allowing users to seamlessly switch between the traditional pointer and the Bubble Cursor for direct comparison during the study.

Both cursor modes (Bubble and Traditional) shared the same interface, visual components, and metric-logging logic (e.g. TrialData class, total cursor distance, selections, deselections, time per trial), ensuring that comparisons between techniques were both fair and precisely measurable.



Performance & Evaluation:

To evaluate the bubble cursor's performance, I conducted a **user study with 6 participants**, each completing a series of Fitts-based tasks (20 total) using both the traditional cursor and my custom bubble cursor. Several key performance metrics were recorded to assess the efficiency and accuracy of each technique:

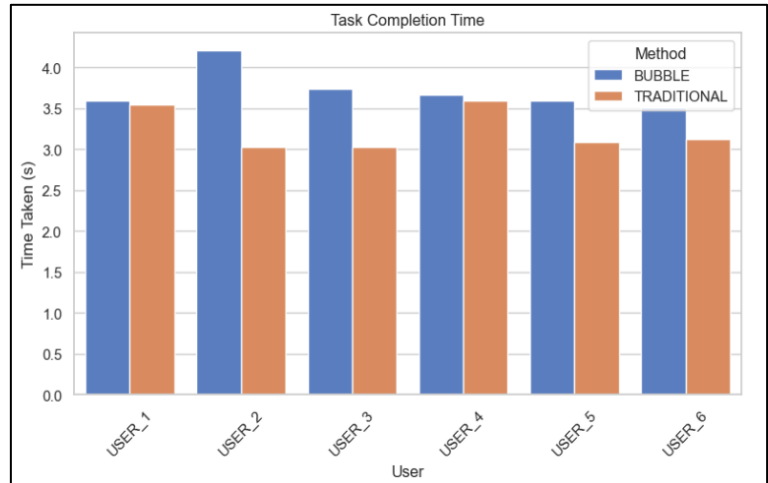
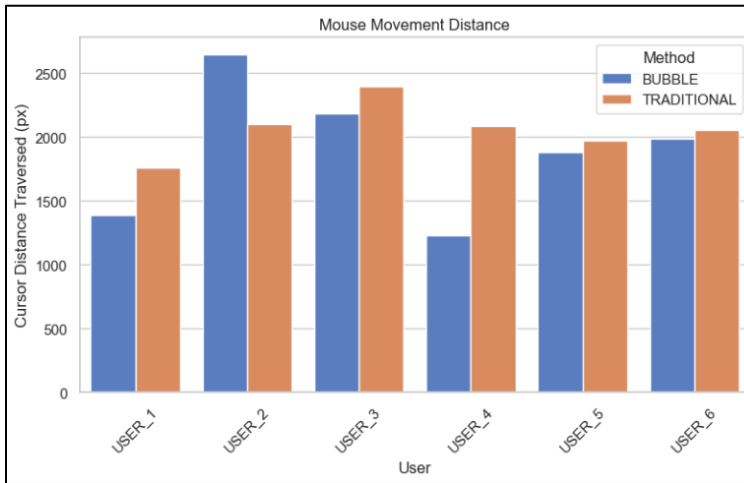
- **Time Taken per Trial:** The total duration (in seconds) required to complete a trial.
 - **Cursor Distance Traveled:** The cumulative distance (in pixels) moved by the cursor during each trial.
 - **Total Selections:** The total number of selections (both correct and incorrect) made.
 - **Total Deselection Rate:** The percentage of selections that were later deselected.
 - **Target Deselection Rate:** The percentage of initially correct target selections that were mistakenly deselected.
 - **Net Accuracy:** The percentage of targets correctly selected out of the required number of targets.
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Participants generally expressed positive sentiments toward the Bubble Cursor:

- *“The attractor curve feature of the bubble cursor was both precise and helpful at guiding my selection.”*
- *“The auto-shading made it clear to me on which item I was going to select”*
- *“At first it was weird seeing the bubble grow, but after a few trials it felt very natural and kind of neat actually.”*

Metrics and other analytics graphs depicting performance shown on the next page*

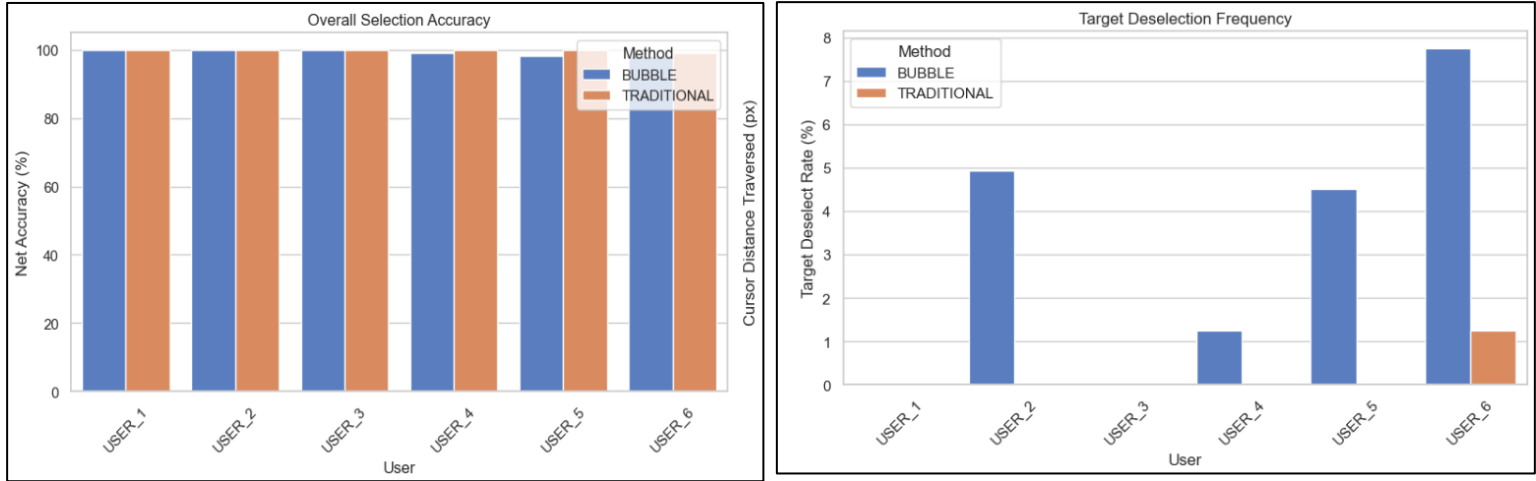
Average [Time Taken (s) | Mouse Distance (px)] per User by Cursor Across 20 Trials:



Immediately, we can observe that the bubble cursor, on average and for most users except USER_2, required less mouse movement than the traditional cursor. This is expected, as the bubble automatically expands to the nearest target, reducing the need for users to move the cursor directly over distant targets. Conversely, we note that the average time taken to complete each trial was slightly higher with the bubble cursor. This suggests a tradeoff between distance and time: although the bubble cursor reduced overall movement, it demanded more precision, especially when users needed to navigate around non-targets to reach a nested target. However, the difference in time was relatively small — ranging from 0.1 to 0.6 seconds, with a maximum difference of about 1 second observed for USER_2.

More analysis with graphics shown on the next page*

Average [Overall Accuracy / Target Deselection] per User by Cursor across 20 trials:



When examining the overall selection and deselection metrics, several interesting trends emerge. Both the traditional mouse and bubble cursor achieved an overall target selection accuracy of essentially 100%, meaning that users consistently selected all the required targets (and occasionally more) during each trial. However, as mentioned earlier, while the bubble cursor minimized the distance users needed to move, the clustering of nearby targets sometimes led to the unintended selection of non-targets. This, in turn, prompted users to perform additional deselections before finalizing their selections. This pattern was particularly noticeable among USERS 2, 4, 5, and 6.

Conclusion:

Overall, the results of this study demonstrate that there are both strengths and tradeoffs when using the bubble cursor as a pointing technique in comparison to the traditional mouse cursor. The bubble cursor successfully reduced the amount of cursor movement required to select targets, validating its intended advantage in minimizing mouse effort. However, the slight increase in completion times suggests that users had to be more deliberate, especially when

navigating densely packed areas to avoid accidentally selecting non-targets. In fact users had to go around non-targets in some cases to select a target or get as close as when using a traditional mouse pointer. Despite these challenges, both techniques achieved nearly perfect overall selection accuracy, with the bubble cursor requiring slightly more corrections through deselections in the clustered environments. These results suggest that while the bubble cursor improves efficiency in terms of distance traveled, the target layout plays a heavier role in fully maximizing its benefits without introducing selection errors. Future iterations and improvements could see the use of additional user input for cluster selection (zoom in on subsection of targets) in addition to the dynamic behavior of the bubble for better accuracy when dealing with nearby distractors.